



Molecular monitoring of CML : BCR-ABL1 detection via GeneXpert in East Algeria Hemobiology laboratory CHU Constantine

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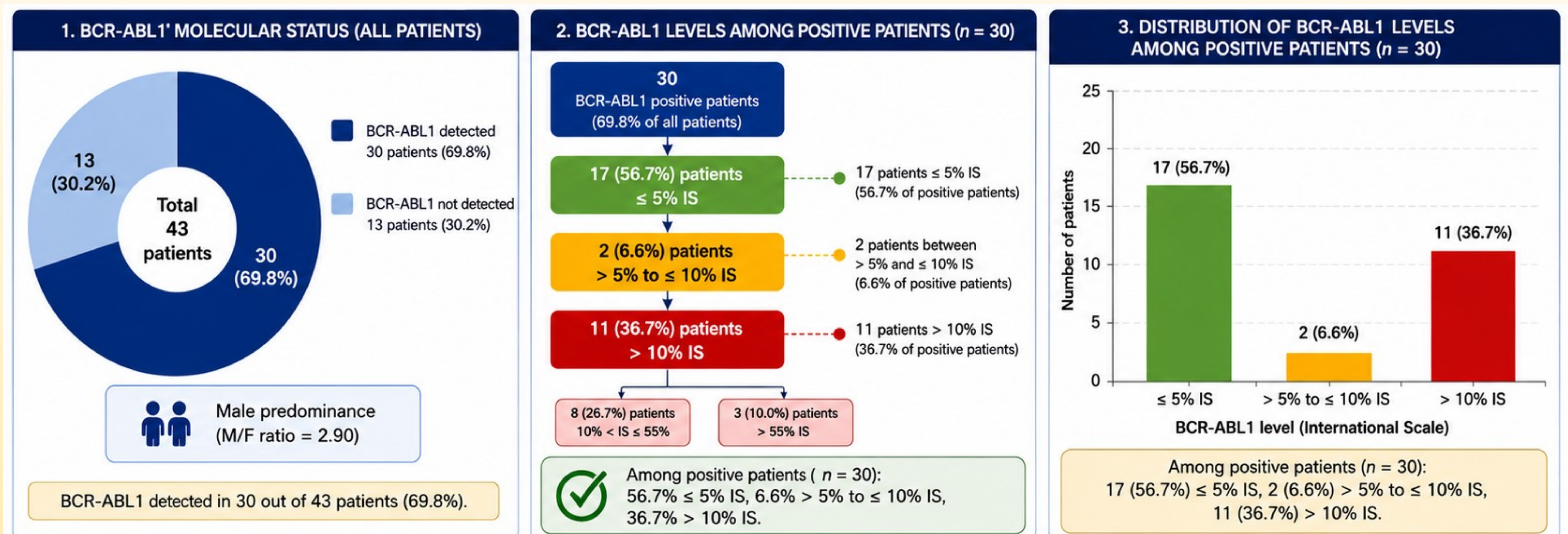
I-OBJECTIVE: To assess the molecular status of patients with CML using the GeneXpert automated real-time PCR platform for the detection and quantification of major BCR-ABL1 transcripts. .

II-INTRODUCTION: Chronic myeloid leukemia (CML) is a myeloproliferative neoplasm characterized by the presence of the BCR-ABL1 fusion gene. Molecular monitoring of BCR-ABL1 transcripts is essential for assessing treatment response and guiding therapeutic management. Quantitative real-time PCR is the reference approach for molecular response assessment and allows BCR-ABL1 transcript levels to be expressed on the International Scale (IS).

II-Materiels and methods : A clinical study was conducted from february 1st to june 28th .blood samples were analyzed using the GeneXpert system , specifically targeting the major breakpoints e13a2/b2a2 and e14a2/b3a2 via automated real-time PCR .The study included 43 patients primarily from Eastern Algeria . the cohort exhibited a notable male predominance with a sex ratio (M/F) of 2.90.

III-Results : Among the **43 patients analyzed**, BCR-ABL1 transcripts were detected in **30 patients (69.8%)**, whereas **13 patients (30.2%)** had no detectable transcript. The cohort showed a marked **male predominance**, with a male-to-female ratio of **2.90**. Most patients were receiving **imatinib**.

Among the 30 patients with detectable BCR-ABL1 transcripts, **13 (43.3%)** had transcript levels above **5% IS**, while **11 (36.7%)** had levels above **10% IS**.



IV-DISCUSSION: The detection of BCR-ABL1 transcripts in 69.8% of patients highlights the importance of molecular monitoring in CML. The observed distribution of BCR-ABL1 levels indicates heterogeneity in molecular response among the studied patients. The GeneXpert platform offers an automated approach that may facilitate molecular testing and improve access to BCR-ABL1 monitoring in regional healthcare settings. (1,2,3,4)

However, the limited sample size and the absence of longitudinal follow-up and comparison with a reference molecular method should be considered when interpreting these findings.

V-CONCLUSION

GeneXpert-based BCR-ABL1 real-time PCR provides a practical and standardized approach for molecular monitoring of CML. Its implementation may facilitate access to molecular response assessment in regional laboratories. Further studies involving larger cohorts and longitudinal follow-up are warranted to better assess its performance in CML treatment monitoring.

KEYWORDS

Chronic myeloid leukemia – BCR-ABL1 – GeneXpert – Real-time PCR – Molecular monitoring – International Scale

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